Set of Exercises

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The code and the LAtex version of this report can be found at https://github.com/AlbanSMA/Comp

1 Introduction

1.1 Exercise 1

The Taylor expansion of a function is :

$$\sum^{n} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Expanding around $y(t_0) = y_0$ and $y(t_1) = y_1$, we get:

$$y(t) = y_0 + \frac{dy_0}{dt}(t - t_0)$$

And:

$$y(t) = y_1 + \frac{dy_1}{dt}(t - t_1)$$

So, if we rewrite these equations, we obtain:

$$\begin{aligned} y(t) &= y_0 + \frac{dy_0}{dt}(t - t_0) \\ \Leftrightarrow y(t) - y_0 &= f(y_0, t_0)(t - t_0) \\ \Leftrightarrow \frac{y_1 - y_0}{\Delta t} &= f(y_0, t_0) \end{aligned}$$

Similarly:

$$\begin{aligned} y(t) &= y_1 + \frac{dy_1}{dt}(t - t_1) \\ \Leftrightarrow y(t) - y_1 &= f(y_1, t_1)(t - t_1) \\ \Leftrightarrow \frac{y_1 - y_0}{\Delta t} &= f(y_1, t_1) \end{aligned}$$

1.2 Exercise 2

see Code/Exercises/Exercise2.py for the Code

We can see in figure **??** and **??** that for the forward Euler method, the longer time steps produce functions which do not fit what is expected at all, walking up and down around the expected function with quite large errors (althought that error is

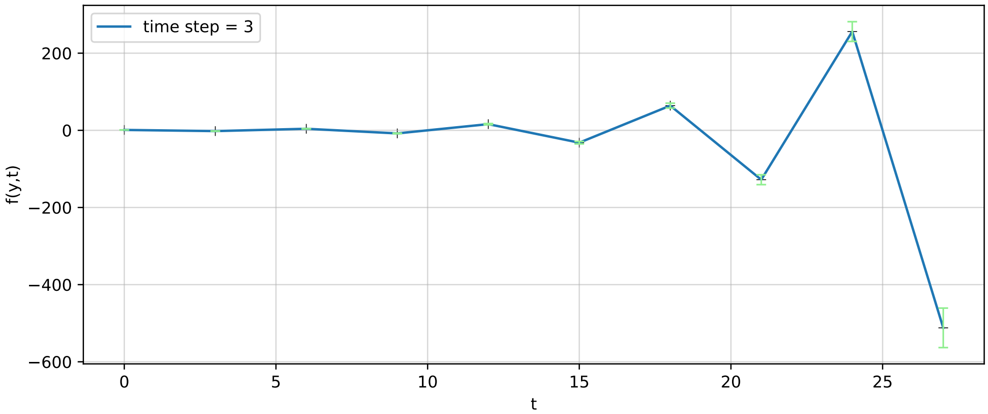

Figure 1: Figure showing the function $\frac{dy}{dt} = -y$ analysed numerically using the forward, or explicit, Euler method with a time step $\Delta t = 3$.

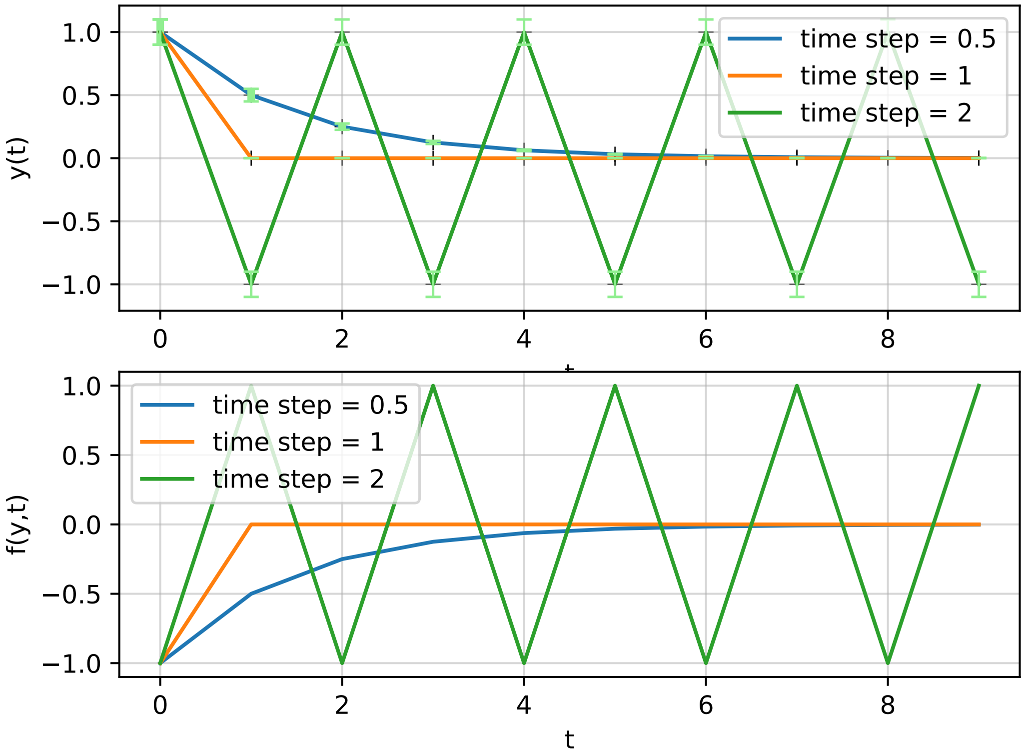

Figure 2: Figure showing the function $\frac{dy}{dt} = -y$ analysed numerically using the forward, or explicit, Euler method for three different time steps. The upper pannel is the function and the lower pannel, its slope.

extremely dependent on the keyboard defined initial error). The shorter time steps however produced a much more reasonable function, and we can see that at $\Delta t = 0.5$, we can already see the real function appear. The initially quite large keyboard defined error is reduced to a very small error within only a few steps.

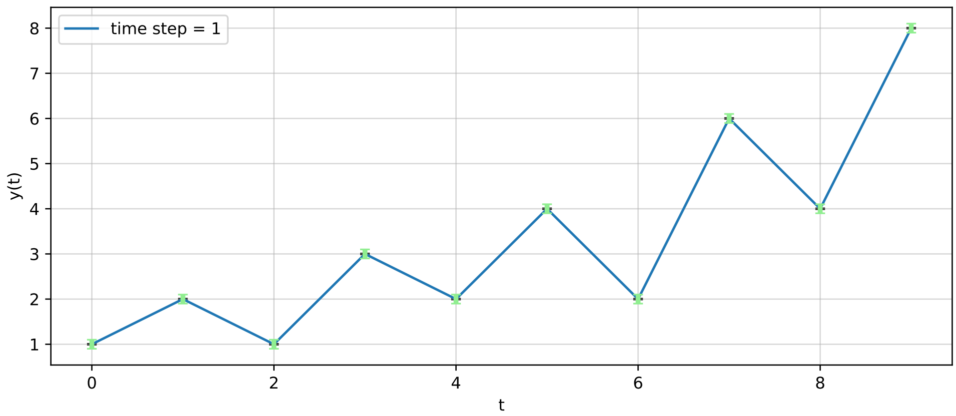

Figure 3: Figure showing the function $\frac{dy}{dt} = -y$, $y(0) = 1$, analysed numerically using the backward, or implicit Euler method and the Newton-Raphson method, with three different time steps. The upper pannel is the function and the lower pannel, its slope. The errorbars are shown in light green.

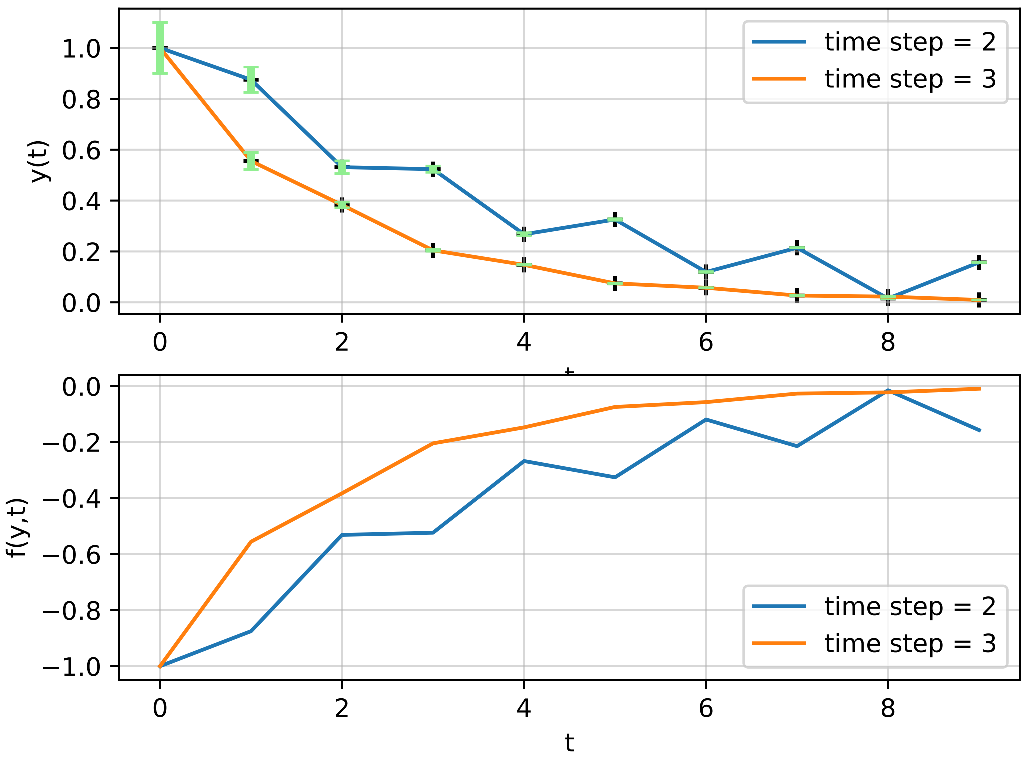

Figure 4: Figure showing the function $\frac{dy}{dt} = -y$, $y(0) = 1$, analysed numerically using the backward, or implicit Euler method and the Newton-Raphson method, with three different time steps. The upper pannel is the function and the lower pannel, its slope. The errorbars are shown in light green.

On the contrary, as we can see on figure **??** and **??**, the implicit method seems to yield better results for longer times steps. For $\Delta t = 1$, the function is increasing when it should be decreasing, while for $\Delta t = 3$, it follows the expected curve. Even for quite high initially keyboard defined error, it falls to much smaller numbers quite fast.

1.3 Exercise 3

Reusing the code from Exercise II, plus the code in Code/Exercises/Exercise3.py

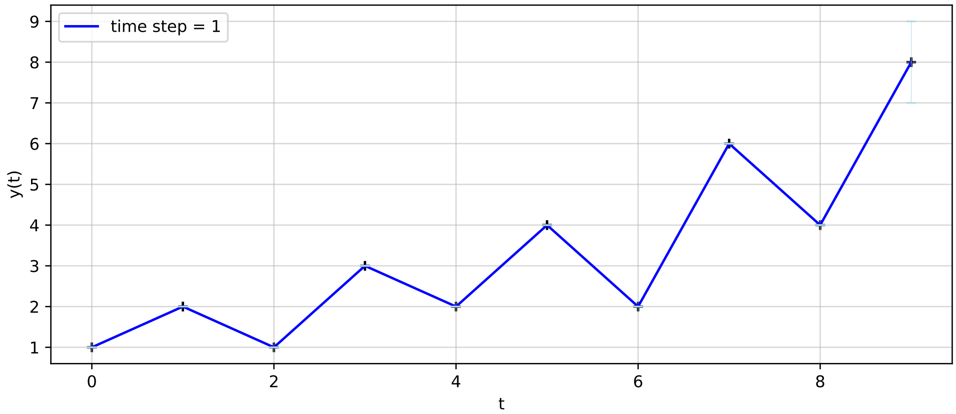

Figure 5: Figure showing the function $\frac{dy}{dt} = \frac{1}{(y^2+1)}$, $y(0) = 1$, analysed numerically using the backward, or implicit Euler method and the Newton-Raphson method, with three different time steps. The upper pannel is the function and the lower pannel, its slope. The errorbars shown in light green are the relative errors from one time step to the next.

For this new function, the longer time steps are once again more accurate than the shorter ones, as we can see on figure **??** and **??**. For $\Delta t = 1$, the function increases when it should be decreasing, although the error remains quite small until the last step where it explodes. On the other hand, for $\Delta t = 2$ and 3, the functions are quite accurate, $\Delta t = 3$ being the most accurate, but the error explodes much faster, and in about equivalent value as for $\Delta t = 1$. The accuracy for any time step size also depends heavily on the initial guess for the value of the second point.

1.4 Exercise 4

To implement a centred derivative, we could use a different method to find a new point away from the initial value, then use this new point to implement the method, then create another one further away with the first method. The projected one does not have to be extremely accurate, as it will be recalculated in the next step. However, if it is too inacurate, the error will only keep growing, as following points are based in part on this one.

Another way of doing this would be to project a new point extremely far with another method, then find a point in the middle, then keep using the centred method on these new points until sufficient precision is reached. The problem here would be the accuracy of the first point, as the step would have to be extremely big, and thus

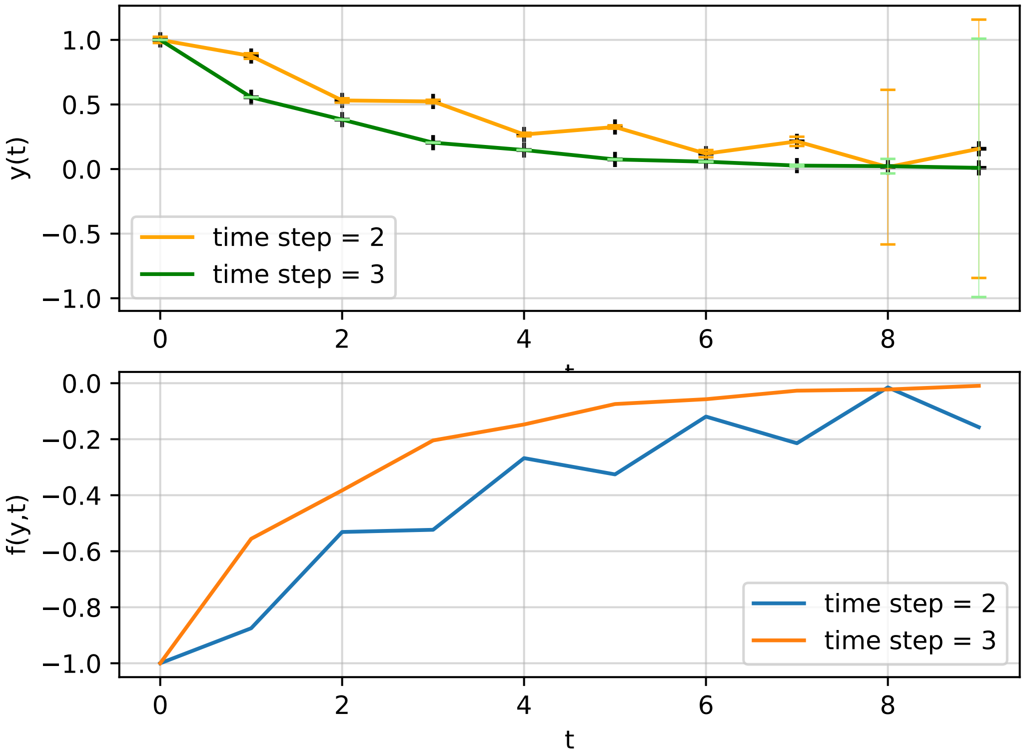

Figure 6: Figure showing the function $\frac{dy}{dt} = \frac{1}{(y^2+1)}$, $y(0) = 1$, analysed numerically using the backward, or implicit Euler method and the Newton-Raphson method, with three different time steps. The upper pannel is the function and the lower pannel, its slope. The errorbars shown in light green are the relative errors from one time step to the next.

not extremely accurate. As the accuracy of the other points depend in part on it, it might cause some larger accuracy further down the line.

2 Nuclear Reaction Networks

2.1 Exercise 5

Reaction 8 :

$$^1_1p + ^1_1p \to ^2_1d$$

It is missing two particles : a positron for the charge, as we have two positive charges on one size and only one on the other side. This however creates an imbalance in lepton number, and so we must add a particle without charge : an electron neutrino. The relation becomes:

$$^1_1p + ^1_1p \to ^2_1d + e^+ + \nu_e.$$

Reaction 9 :

$$^1_1p + ^7_3Li \to 2^4_2He$$

This equation is mostly balanced : there are 8 nucleons and 4 charges on both sides, but it is missing a lepton on the right hand side. The relation becomes :

$$ {}^{1}_{1}p + {}^{7}_{3}Li \rightarrow 2{}^{4}_{2}He + \nu_e $$

Reaction 12 :

$$ 3{}^{4}_{2}He \rightarrow {}^{12}_{6}C $$

This reaction is already balanced : there are 12 nucleons, 6 charges and no leptons on both sides.

2.2 Exercise 6

$$
\begin{aligned}
\frac{dn_{^2He}}{dt} &= r_1 n_p^2 - r_2 n_{^2He} n_p \\
\frac{dn_{^3He}}{dt} &= r_2 n_{^2He} n_p - r_3 n_{^3He}^2 \\
\frac{dn_{^4He}}{dt} &= r_3 n_{^3He}^2 \\
\frac{dn_p}{dt} &= -r_1 n_p^2 - r_2 n_p n_{^2He} + r_3 n_{^3He}^2
\end{aligned} \tag{1}
$$

Where r_1, the reaction rate for reaction 17 is: $10^{-43}\ cm^3/s$, r_2, the reaction rate for relation 18 is: $10^{-26}\ cm^3/s$ and r_3, the reaction rate for relation 19 is: $10^{-34}\ cm^3/s$.

To find the jacobian, we need to determine each of its elements:

$$ J_{ij} = \frac{\partial \tilde{y}_i}{\partial y_j} $$

Where $\tilde{y}_i$ is the function $\frac{y_{i,1} - y_{i,0}}{dt} - f(y_{i,1}, t_{i,1})$. In this case, y is the number density of one of the elements. Let us start with taking $y \rightarrow n_{^2He}$:

$$
\begin{aligned}
\tilde{n}_{^2He} &= \frac{n_{^2He,1} - n_{^2He,0}}{dt} - r_1 n_p^2 + r_2 n_{^2He,1} n_p \\
\Leftrightarrow \frac{\partial \tilde{n}_{^2He}}{\partial n_{^3He}} &= 0 \\
\Leftrightarrow \frac{\partial \tilde{n}_{^2He}}{\partial n_{^4He}} &= 0 \\
\Leftrightarrow \frac{\partial \tilde{n}_{^2He}}{\partial n_p} &= -2r_1 n_p + r_2 n_{^2He,1} \\
\Leftrightarrow \frac{\partial \tilde{n}_{^2He}}{\partial n_{^2He}} &= \frac{1 - n_{^2He,0}}{dt} + r_2 n_p
\end{aligned}
$$

Then for $n_{^3He}$:

$$\begin{aligned}
\tilde{n}_{^3He} &= \frac{n_{^3He,1} - n_{^3He,0}}{dt} - r_2 n_{^2He} n_p + r_3 n^2_{^3He,1} \\
\Leftrightarrow \frac{\partial \tilde{n}_{^3He}}{\partial n_{^2He}} &= -r_2 n_p \\
\Leftrightarrow \frac{\partial \tilde{n}_{^3He}}{\partial n_{^4He}} &= 0 \\
\Leftrightarrow \frac{\partial \tilde{n}_{^3He}}{\partial n_p} &= -r_2 n_{^2He} \\
\Leftrightarrow \frac{\partial \tilde{n}_{^3He}}{\partial n_{^3He}} &= \frac{1 - n_{^3He,0}}{dt} + 2r_3 n_{^3He,1}
\end{aligned}$$

And $n_{^4He}$

$$\begin{aligned}
\tilde{n}_{^4He} &= \frac{n_{^4He,1} - n_{^4He,0}}{dt} - r_3 n^2_{^3He} \\
\Leftrightarrow \frac{\partial \tilde{n}_{^4He}}{\partial n_{^2He}} &= 0 \\
\Leftrightarrow \frac{\partial \tilde{n}_{^4He}}{\partial n_{^3He}} &= -2r_3 n_{^3He} \\
\Leftrightarrow \frac{\partial \tilde{n}_{^4He}}{\partial n_p} &= 0 \\
\Leftrightarrow \frac{\partial \tilde{n}_{^4He}}{\partial n_{^4He}} &= \frac{1 - n_{^4He,0}}{dt}
\end{aligned}$$

Finally for n_p:

$$\begin{aligned}
\tilde{n}_p &= \frac{n_{p,1} - n_{p,0}}{dt} + r_1 n_p^2 + r_2 n_p n_{^2He} - r_3 n^2_{^3He} \\
\Leftrightarrow \frac{\partial \tilde{n}_p}{\partial n_{^2He}} &= r_2 n_p \\
\Leftrightarrow \frac{\partial \tilde{n}_p}{\partial n_{^3He}} &= -2r_3 n_{^3He} \\
\Leftrightarrow \frac{\partial \tilde{n}_p}{\partial n_{^4He}} &= 0 \\
\Leftrightarrow \frac{\partial \tilde{n}_p}{\partial n_p} &= \frac{1 - n_{p,0}}{dt} + 2r_1 n_p + r_2 n_{^2He}
\end{aligned}$$

So the complete jacobian is :

$$\begin{bmatrix}
\frac{1-n_{p,0}}{dt} + 2r_1 n_p + r_2 n_{^2He} & r_2 n_p & -2r_3 n_{^3He} & 0 \\
-2r_1 n_p + r_2 n_{^2He,1} & \frac{1-n_{^2He,0}}{dt} + r_2 n_p & 0 & 0 \\
-r_2 n_{^2He} & -r_2 n_p & \frac{1-n_{^3He,0}}{dt} + 2r_3 n_{^3He,1} & 0 \\
0 & 0 & -2r_3 n_{^3He} & \frac{1-n_{^4He,0}}{dt}
\end{bmatrix} \tag{2}$$

And we are excluding the 1/dt terms, so we find:

$$\begin{bmatrix}
2r_1 n_p + r_2 n_{^2He} & r_2 n_p & -2r_3 n_{^3He} & 0 \\
-2r_1 n_p + r_2 n_{^2He} & r_2 n_p & 0 & 0 \\
-r_2 n_{^2He} & -r_2 n_p & 2r_3 n_{^3He} & 0 \\
0 & 0 & -2r_3 n_{^3He} & 0
\end{bmatrix} \tag{3}$$

To find the stiffness of the system, we need to calculate its eigenvalues. The code used to do so can be found in *Code/Exercises/SysStiffness.py*. There seem to be an error in the calculation of the Jacobian, as the eigenvalues seem strange. However, by playing around with the densities in this system, and with a fixed number densities for the protons, we can make some observations. The stiffness varies most with the density of deuterium, and least with ^4He. It is stiffer when these densities are lower, and less stiff when they are higher. If the densities are too low, the eigenvalues are complex and it is more complicated to calculate the stiffness.

2.3 Exercise 7

a. There are 4 types of particles, and so 4 equations are needed to describe the system.

b. The system of equations is :

$$
\begin{aligned}
\frac{dn_{Cr}}{dt} &= -r_1 n_{Cr} n_\alpha \\
\frac{dn_{Fe}}{dt} &= r_1 n_{Cr} n_\alpha - r_2 n_{Fe} n_\alpha \\
\frac{dn_{Ni}}{dt} &= r_2 n_{Fe} n_\alpha \\
\frac{dn_\alpha}{dt} &= -r_1 n_{Cr} n_\alpha - r_2 n_{Fe} n_\alpha
\end{aligned}
$$

So the Jacobian is :

$$
\begin{bmatrix}
-r_1 n_{Cr} - r_2 n_{Fe} & -r_1 n_\alpha & -r_2 n_{Fe} & 0 \\
-r_1 n_{Cr} & -r_1 n_\alpha & 0 & 0 \\
r_1 n_{Cr} - r_2 n_{Fe} & r_1 n_\alpha & -r_2 n_\alpha & 0 \\
r_2 n_{Fe} & 0 & r_2 n_\alpha & 0
\end{bmatrix}
$$

c. If we add the radioactive decay of ^{56}Ni to the system, then we need to add terms to most of the equations, and thus the terms of the Jacobian. The ^{56}Ni might decay to ^{52}Fe, also releasing α particles in the process. Then there needs to be new destructive terms in the ^{56}Ni terms, and new constructive ones in the ^{52}Fe and α ones. Depending on what the ^{56}Ni decays to, there might also be a new equation in the system, representing a new particle. This particle would only have constructive terms.

Similarly, photodissociation would add new destructive terms to the equations already in the system, and possibly constructive terms to new equations. This would depend on what the particles already present decay to.

Finally, three-body reactions would also add both constructive and destructive terms to the already present equations, and create new ones, thus also adding terms to the Jacobian, and new rows and columns. For example, the three-body reaction $3\alpha \rightarrow^{12} C$ would add a destructive term to the alpha equation, and thus also in the first term of the Jacobian (term J_{11}). It would also create a new equation, and a new row and collumn with a constructive term to add the presence of ^{12}C to the system.